

PMDD

Prediction, Monitoring & Dockerized Deployment

EVA Machine Learning Reasoning Layer — Module Specification

Version 1.0 | References: EVA-ML-Master.docx

1. Purpose

The Prediction, Monitoring & Dockerized Deployment Module converts a validated model into a reusable prediction service and continuously evaluates whether its predictions remain reliable over time. This module allows EVA to move from analysis to operational assistance.

Core Principle

A trained model is not a finished product. Reality changes — user behavior shifts, market conditions move, data patterns drift. EVA does not simply deploy models. It supervises them. The module treats the model as a monitored instrument, not a permanent truth source.

2. Preconditions

This module activates only if the RVEM has assigned a validation status of Validated or Conditionally Validated. Rejected models cannot be deployed under any circumstances.

⚠ Hard Rule

A model with Rejected validation status cannot be deployed, cannot serve predictions, and cannot be presented to the user as a result. This rule has no exceptions.

3. Dockerized Model Service

The selected model is packaged as an isolated containerized service. The purpose of containerization is reproducibility, independence from the main EVA system, consistent behavior across machines, and safe execution.

The container holds the trained model, feature schema, preprocessing rules, prediction logic, and explanation interface. The model must never rely on external state at prediction time.

4. Prediction Interface

The deployed service accepts new input records that follow the Learning View schema. For each record, the system returns:

- Predicted outcome
- Confidence estimate (mandatory — predictions without confidence are not permitted)
- Risk category
- Explanation reference pointing to the RVEM case-level explanation

For Conditionally Validated models, all predictions are accompanied by a caution indicator in the response payload and in the dashboard.

5. Feature Consistency Check

Before generating a prediction, the module verifies that required features are present, feature formats are valid, and values fall within expected ranges. If input is inconsistent, the system must either reject the prediction with a logged reason, or serve the result flagged as unreliable with an explanation.

6. Prediction Logging

Every prediction request is recorded in the `prediction_logs` session artifact. The log includes timestamp, input summary, prediction, confidence level, and any flags raised during feature consistency check. This log supports monitoring and later review.

7. Drift Monitoring

The module continuously checks for model degradation using the thresholds defined in the Master PRD. The following drift types are monitored:

Drift Type	Alert Threshold
Confidence Drift	Average prediction confidence drops more than 15% relative to baseline over a rolling 100-prediction window.
Data Drift	Input feature distribution diverges from training distribution beyond a PSI threshold of 0.2 for any key feature.
Behavior Drift	Distribution of predicted outcomes shifts more than 20% relative to the baseline prediction distribution.
Error Pattern Drift	If ground truth is available, error rate exceeds 2x the holdout test error rate over any 200-prediction window.

These thresholds are defaults and may be adjusted by the analyst in the session configuration before deployment.

For Conditionally Validated models, monitoring runs at 2x the standard frequency for all drift types.

8. Retraining Recommendation

When drift thresholds are exceeded, the module recommends retraining by writing a retraining flag to the `prediction_deployment_record` and alerting the Dashboard Composer. The system does not automatically retrain. A new analysis session must be initiated by a human reviewer.

⚠ No Automatic Retraining

EVA never retrains a model without human review. An alert is raised and a new session is requested. This is by design — automatic retraining without review could silently degrade the quality of decisions.

9. Human Review Touchpoints

Touchpoint	Behavior
Drift Alert	Dashboard displays alert. Human must acknowledge and decide whether to initiate a new session.
Conditionally Validated Predictions	For high-stakes decisions, a human must log acknowledgement before acting on predictions.
Model Override	An analyst may override a specific prediction. The override is logged in the session_log with reason.
Retraining Decision	Human must initiate new session. System does not retrain autonomously.

10. Integration with Dashboard Composer

The Dashboard Composer receives high-risk predictions, uncertain cases, drift alerts, and Conditionally Validated model indicators. The dashboard must visibly indicate when model reliability is weakening. For conditionally validated models, a persistent orange caution banner is shown on all predictions.

11. Integration with Report Generator

The Report Generator includes how predictions are produced, reliability limitations, current monitoring status, and any retraining recommendations. The report must never present predictions without explaining uncertainty. For Conditionally Validated models, the report must include a dedicated section on unresolved conditions and associated risks.

12. Output: prediction_deployment_record

Field	Contents
Deployment Status	Model active or inactive
Validation Status	Validated or Conditionally Validated
Prediction Schema	Required input features and formats
Confidence Behavior	Typical prediction confidence profile at deployment
Monitoring Results	Detected drift types and current status
Retraining Recommendation	Flag and reason if triggered

13. Operational Rules

1. Only Validated or Conditionally Validated models may be deployed.

2. Every prediction must include a confidence estimate.
3. Every prediction must be logged with timestamp and input summary.
4. Drift must be monitored continuously using the defined thresholds.
5. Users must be warned when model reliability is declining.
6. Automatic retraining is prohibited. Human review is required.
7. Conditionally Validated models run monitoring at 2x standard frequency.
8. Model overrides must be logged with reason.

14. Relationship to Other Modules

Module / Component	Relationship
RVEM	Approves model and provides validation status
Dashboard Composer	Receives predictions, alerts, and caution indicators
Report Generator	Receives monitoring status and limitation explanations
GAL (session_log)	Stores full monitoring history and override log
Master PRD	Defines drift thresholds, conditional validation behavior, and retraining rules